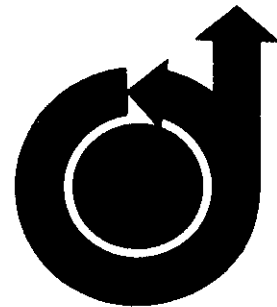


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**A THEORETICAL STUDY OF NONLINEAR DAMPING
BY HELMHOLTZ RESONATORS**

by

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A THEORETICAL STUDY OF NONLINEAR DAMPING
BY HELMHOLTZ RESONATORS*

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Abstract

A theoretical study of the interaction between finite-amplitude sound and a single Helmholtz resonator is hereby presented. The flows in the entrance region, the orifice and the cavity have been considered in detail with the aid of the conservation equations. To account for the nonlinearities of the problem, the differential equations describing first - as well as second - order effects have been solved. The results of this study indicate that the experimentally-observed losses can be attributed to the following two mechanisms: (1) Viscous damping and (2) Energy loss which is associated with the dissipation of the kinetic energy of the jets which are periodically formed at both ends of the orifice; the latter is amplitude-dependent. Plots of the theoretically-predicted resistance (which includes both viscous and "jet" losses) as functions of velocity amplitude are in excellent agreement with available experimental data. Typical plots of the resonator admittance and phase angle between pressure oscillations on both sides of the orifice vs. frequency are presented.

Introduction

The interest in this problem has been renewed in recent years because the use of an array of Helmholtz resonators proved successful in suppressing combustion instability in liquid-propellant rocket motors (e.g., see ref. 1). Arrays of Helmholtz resonators (the so-called acoustic liners) have also been used in the reduction of high-intensity noise⁽²⁾ and the elimination of combustion instability in air-breathing engines⁽³⁾. In order to develop a rational design procedure for acoustic liners it is important to understand the behavior of the flow in the vicinity of the resonators when they are subjected to various oscillatory conditions; in particular it is important to understand the nonlinear interaction between the resonators and the flow when the amplitude of the oscillation is large. In an effort to achieve this goal the present paper presents a theoretical study of the nonlinear response of a single Helmholtz resonator that is subjected to finite-amplitude pressure oscillations.

The linear theory of Helmholtz resonators was treated mathematically for the first time by Helmholtz in 1860 and was afterward greatly simplified by Lord Rayleigh (1871) (see ref. 4 for an account of early studies). Since that time considerable effort has been expended on studies directed towards obtaining better understanding of these mechanical devices. Due to mathematical and experimental difficulties as well as insufficient motivation the

majority of these studies was limited to the linear domain, where finite-amplitude may be neglected.

Nonlinear studies of Helmholtz resonators are scarce in number and relatively recent. Important among these are the experimental investigations of Ingard^(5,6,7), Blackman⁽⁸⁾, Bies and Wilson⁽⁹⁾, Garrison⁽¹⁰⁾ as well as the theoretical work of Sirignano⁽¹¹⁾. One of the important results of these investigations is the observation that the resistance of a Helmholtz resonator increases as the amplitude of the pressure oscillation increases. Ingard⁽⁵⁾ and Sirignano⁽¹¹⁾, in an effort to explain these observations, rightly suggested that the observed increase in the resonator resistance is due to the dissipation into turbulence of the kinetic energy of the jets that are periodically formed at both ends of the resonator's orifice. In the present paper a more complete description of the flow field is presented, some of the early assumptions are removed, and the theory is accordingly modified.

Description of Resonator Flow Field

Existing experimental evidence^(5,7) indicates that the flow field in the vicinity of an orifice that is subjected to a periodic pressure oscillation is very complex; it depends on the frequency and amplitude of the oscillation as well as the geometry of the orifice (i.e., l/d -its length to diameter ratio). It is shown in ref. 5 that for a given orifice geometry and frequency of oscillation, secondary flow patterns which resemble vortex rings and are stationary relative to the orifice can be established at relatively low amplitudes. An increase in the amplitude of the oscillation results in a change in the direction of rotation of the secondary flow patterns; a further increase in amplitude results in the appearance of jets (that co-exist with the secondary vortex motion) at the orifice's exit; a further increase in amplitude results in the "strengthening" of the jets and the disappearance of the secondary flow patterns. In the region of operation where the periodic jet pulses consistently appear at the orifice's exit the flow into the orifice appears⁽¹²⁾ to be radial in nature and the direction of the leaving jet is axial (see Fig. 1). The shear present at the jet's boundaries results in the distortion of the jet and the formation of a vortex ring. While the vortex ring is being formed, the pressures on both sides of the orifice change and the flow in the orifice must change direction. Correspondingly the character of the flow must change with the incoming flow, that is moving around the just-formed vortex ring, entering the orifice from an almost radial direction and leaving on the other side as an axially-directed jet. This flow pattern reverses itself twice during each cycle of oscillation. Once formed, the vortex rings move away from the orifice plate and in the process they disintegrate into turbulence. The formation and disintegration of the vortex ring appear to be the major energy-

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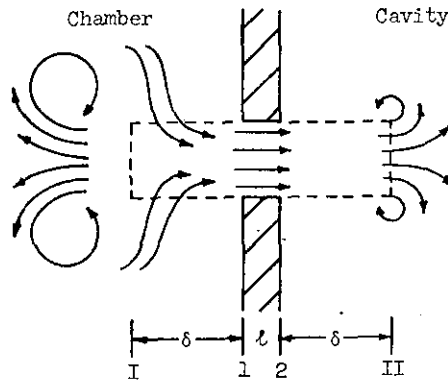


Figure 1. Schematic of Resonator Flow Field

dissipation mechanisms; and, as suggested previously^(5,11), they must be responsible for the observed increase in resonator losses.

Analysis

The nonlinear analysis herein presented is primarily aimed at obtaining a better understanding of the resonator behavior in the region of operation in which the pulsating jets are formed. Before proceeding with the analysis some of the basic assumptions used in this work will be stated; they are: (1) The gas is calorically perfect; (2) There is no mean flow through the orifice; (3) The flow in the orifice and in the vicinity of its exit is one-dimensional; (4) The flow in the orifice is not choked at any time; (5) The flow conditions on both sides of the orifice are symmetrical and they exactly reverse themselves during the cycle (that is, inflow and outflow conditions on both sides of the orifice's plate are identical at corresponding instances of the cycle); (6) The flow at exit plane is always separated; (7) At each instant the amplitude of the axial component of the velocity remains constant along the orifice and the jet; (8) Viscous forces are of second order; and (9) The axial momentum of the flow entering the side boundaries of the control volume in front of the orifice may be neglected.

To obtain a solution the following conservation equations for the perturbations that are correct to second order will be required:

$$\frac{d}{dt} \int_V \rho' dV = - \int_S (\bar{\rho} \mathbf{v}' + \rho' \mathbf{v}) \cdot \mathbf{n} dS \quad (1)$$

$$\frac{d}{dt} \int_V (\bar{\rho} \mathbf{v}' + \rho' \mathbf{v}) dV = - \int_S [\mathbf{v}' (\bar{\rho} \mathbf{v}' \cdot \mathbf{n}) + p' \mathbf{n}] dS + \{\text{viscous forces}\} \quad (2)$$

$$\frac{d}{dt} \int_V \left(\frac{p'}{\gamma-1} + \frac{1}{2} \bar{\rho} v'^2 \right) dV = - \int_S \left(\frac{\gamma}{\gamma-1} \bar{p} \mathbf{v}' + p' \mathbf{v}' \right) \cdot \mathbf{n} dS \quad (3)$$

In the above equations ρ , p , $\mathbf{v}$ and $\mathbf{n}$ respectively denote the density, pressure, velocity and a unit vector directed away from the surface S . It is important to note that the viscous forces that appear in the momentum equation and are of second order in magnitude are left unspecified. The viscous work terms that result from the presence of viscous forces are of higher order, and they do not appear in the energy equation.

In order to describe the linear and nonlinear aspects of the gas oscillations in the resonator the method of successive approximations⁽¹³⁾ will be used in the analysis. Accordingly each perturbation quantity is assumed to have a series expansion of the following form:

$$p' = p^{(1)} + p^{(2)} + O(\epsilon^3) \quad (4)$$

where the superscripts (1) and (2) respectively denote the first and second order approximations of the solution, ϵ represents the amplitude of the pressure oscillation at section I (see Fig. 1) that is located inside the chamber (i.e., $p_I = \bar{p} + \epsilon \cos \omega t$).

Expanding all the perturbations in the manner described above, substituting these expansions into Eqs. (2) and (3), using some of the previously outlined assumptions, and separating the resulting equations according to their order of magnitude yields the following set of first-order equations for the cylindrical control volumes described in Fig. 1:

$$\bar{\rho} \frac{d}{dt} \int_I u^{(1)} dx = p_I' - p_2^{(1)} \quad ; \quad p_2^{(1)} = p_{II}^{(1)} \quad (5)$$

$$\frac{d}{dt} \int_{V_c} p^{(1)} dV = \gamma A \bar{\rho} u_2^{(1)} \quad (6)$$

In the above equations subscripts denote locations described in Fig. 1; V_c is the cavity volume, A is the area of the orifice cross-section, and u the axial component of the velocity. In deriving the above equations the pressure was assumed to be approximately uniform across the normal faces of the analyzed control volumes. Assuming that $u^{(1)}$ and $p^{(1)}$ can be represented as products of time- and space-dependent quantities (i.e., they are separable), then the time-dependent factors can be removed from inside the integral signs and Eqs. (5) and (6) can be combined to yield

$$\frac{\ell_e V_e}{A \bar{c}^2} \frac{d^2 p_{II}^{(1)}}{dt^2} + p_{II}^{(1)} = \frac{1}{\omega_0^2} \frac{d^2 p_{II}^{(1)}}{dt^2} + p_{II}^{(1)} = \epsilon \cos \omega t \quad (7)$$

where $\bar{c}$ is the steady-state speed of sound in the resonator's cavity, $\omega_0^2 = A \bar{c}^2 / \ell_e V_e$ and for flow from chamber to cavity

$$l_e = \int_I^1 \tilde{u}(x) dx + l \quad ; \quad v_e = \int_{V_c} \tilde{p}(x) dx \quad (8)$$

which represent orifice effective length and cavity effective volume; $\tilde{u}(x)$ and $\tilde{p}(x)$ represent the spatial dependence of the velocity and pressure perturbations. The analytical determination of these quantities would require a more detailed analysis of the flow field in the vicinity of the orifice.

To proceed with the nonlinear analysis the appropriate second-order equations must be derived. The second-order energy equation for the resonator's cavity is:

$$\frac{d}{dt} \int_{V_c} p^{(2)} dV = A_V (\bar{p} u_2^{(2)} + p_2^{(1)} u_2^{(1)}) - (\gamma-1) \cdot \frac{d}{dt} \int_{V_c} \frac{1}{2} \rho v^{(1)2} dV \quad (9)$$

Because of the different natures of the orifice inflow and outflow it becomes necessary to give separate considerations to the second-order momentum equations describing these flows. During half the cycle when the flow is directed from the chamber to the cavity the second-order axial momentum equations can be expressed as follows:

$$\begin{aligned} \frac{d}{dt} \int_I^2 \bar{p} u^{(2)} dx &= p_I' - p_2^{(2)} - \bar{p} u_2^{(1)2} + (\tau/A) \\ &\quad - \frac{d}{dt} \int_I^2 \rho^{(1)} u^{(1)} dx \\ p_2^{(2)} &= p_{II}^{(2)} \end{aligned} \quad (10)$$

and when the flow is from the cavity to the chamber Eq. (10) is replaced by

$$\begin{aligned} \frac{d}{dt} \int_I^{II} \bar{p} u^{(2)} dx &= p_I^{(2)} - p_{II}^{(2)} + \bar{p} u_I^{(1)2} + (\tau/A) \\ &\quad - \frac{d}{dt} \int_{II}^I \rho^{(1)} u^{(1)} dx \\ p_I^{(2)} &= p_I' \end{aligned} \quad (11)$$

In Eqs. (10) and (11) (τ/A) represents an axial viscous force that is exerted on the gas inside the control volume under consideration. Combining Eqs. (10) and (11) and using assumption (7) yields the following second-order momentum equation:

$$\begin{aligned} \bar{p} l_e \frac{du}{dt} &= p_I' - p_{II}^{(2)} - \bar{p} u^{(1)2} + (\tau/A) \\ &\quad - \frac{d}{dt} \int_I^{II} \rho^{(1)} u^{(1)} dx \end{aligned} \quad (12)$$

that applies throughout the duration of the cycle; the minus sign in front of $\bar{p} u^{(1)2}$ holds when the flow is from the chamber to the cavity, and the plus sign holds when the flow is from the cavity to the chamber. The third and fourth terms on the right-hand side of Eq. (12) respectively represent the momentum dissipated by the alternately-formed jets at the orifice's ends and the viscous force exerted on the gas.

To obtain a solution of Eq. (12) it is necessary to specify the various terms which appear in its inhomogeneous part. The solution of Eqs. (6) and (7) yields:

$$u^{(1)} = \frac{\omega_0^2}{\omega^2 - \omega_0^2} \frac{V_e \omega}{\gamma A \bar{p}} e \sin \omega t = \alpha^{(1)} \sin \omega t \quad (13)$$

$$p_{II}^{(1)} = -\alpha^{(1)} \frac{\gamma A \bar{p}}{V_e \omega} \cos \omega t \quad (14)$$

Using Eq. (13) it can be shown by Fourier expansion that

$$\begin{aligned} \mp \bar{p} u^{(1)2} &= \mp \bar{p} \alpha^{(1)2} \sin^2 \omega t = -\frac{8}{3\pi} \bar{p} \alpha^{(1)2} \sin \omega t \\ &\quad + \text{higher harmonics} \end{aligned} \quad (15)$$

According to Ingard⁽⁶⁾ the viscous force per unit area acting on the gas in the orifice and its immediate vicinity can be described by the following empirical relation:

$$(\tau/A) = -\sqrt{8\mu\bar{p}\omega} \left(1 + \frac{l}{d}\right) u^{(1)} \quad (16)$$

Substituting Eqs. (15) and (16) into Eq. (12) yields the following second-order momentum equation:

$$\begin{aligned} \bar{p} l_e \frac{du}{dt} &= p_I' - p_{II}^{(2)} - \left(\frac{8}{3\pi} \bar{p} \alpha^{(1)2} + \sqrt{8\mu\bar{p}\omega} \left(1 + \frac{l}{d}\right)\right) u^{(1)} \\ &\quad - \frac{d}{dt} \int_I^{II} \rho^{(1)} u^{(1)} dx + \text{higher harmonics} \end{aligned} \quad (17)$$

The second-order equations (i.e., Eqs. (9) and (17)) have two kinds of solutions; a particular solution that has the same time dependence as the first-order solution as well as particular solutions that are proportional to higher harmonics. Using the principle of equivalent linearization⁽¹³⁾ in the solution of Eqs. (9) and (17) and following the same procedure as used in the derivation of the first-order equations, it can be shown that the second-order approximation for the cavity pressure that is proportional to the fundamental harmonic (to be denoted by a tilde) is controlled by the following ordinary differential equation:

$$\frac{\ell_e v_e}{Ac^2} \frac{d^2 \tilde{p}_{II}^{(2)}}{dt^2} + \frac{v_e}{\gamma A \bar{p}} \left\{ \left(1 + \frac{\ell}{d}\right) \sqrt{8 \bar{p} \mu \omega} + \frac{8}{3\pi} \bar{p} \alpha^{(2)} \right\} \frac{d \tilde{p}_{II}^{(2)}}{dt} + \tilde{p}_{II}^{(2)} = \tilde{p}_I = \epsilon \cos \omega t \quad (18)$$

where ℓ_e and v_e have definitions similar to those given in Eq. (8) and $\alpha^{(2)}$ is the second-order approximation for the velocity amplitude. The second-order solutions are given by the following expressions:

$$\begin{aligned} \tilde{p}_{II}^{(2)} &= \frac{-\omega_0^2 \bar{p} \ell_e \epsilon \cos(\omega t + \phi)}{\omega \left\{ \left[\omega \ell_e \bar{p} \left(1 - \left(\frac{\omega_0}{\omega}\right)^2\right) \right]^2 + \left[\left(1 + \frac{\ell}{d}\right) \sqrt{8 \bar{p} \mu \omega} + \frac{8}{3\pi} \bar{p} \alpha^{(2)} \right]^2 \right\}^{\frac{1}{2}}} \\ &= -\frac{A \gamma \bar{p}}{V \omega} \alpha^{(2)} \cos(\omega t + \phi) \quad (19) \end{aligned}$$

$$u^{(2)} = \alpha^{(2)} \sin(\omega t + \phi) \quad (20)$$

where

$$\phi = \tan^{-1} \frac{\sqrt{8 \bar{p} \mu \omega} \left(1 + \frac{\ell}{d}\right) + \frac{8}{3\pi} \bar{p} \alpha^{(2)}}{\ell_e \bar{p} \omega \left(1 - \left(\frac{\omega_0}{\omega}\right)^2\right)} \quad (21)$$

The Irrotational Flow Case

Under certain limiting conditions the flow in the entrance region may be assumed to be irrotational. In order to treat this case a different form of the momentum equation must be used. Multiplying the differential version of Eq. (1)

by v' and subtracting the resultant equation from the differential version of Eq. (2) results in the derivation of the Euler equation for the perturbations. Integrating Euler's equation over some as yet unspecified volume V yields the following result:

$$\frac{d}{dt} \int_V \bar{p} v' dv + \int_V \bar{p} (\nabla \chi \nabla) \chi v' dv = - \int_S \left[\frac{1}{2} \bar{p} v'^2 + p' \right] n ds - \int_V \rho' \frac{dv'}{dt} dv \quad (22)$$

In the case of irrotational flow in the entrance region and the orifice the second term on the left-hand side of Eq. (22) is identically zero. Letting this term equal zero, using Eq. (22) instead of Eq. (2), and repeating the analysis of the previous section yields a final result that shows that the lack of rotation in the entrance region reduces the nonlinear resistance by a factor of two. The nonlinear resistance predicted in this case will be identical to the one obtained by Sirignano⁽¹¹⁾.

Results and Discussion

For years it has been customary to analyze the resonator behavior by studying the behavior of a mechanical analog made up of a mass-spring-dashpot combination. Using the relation $u = d\xi/dt$, where ξ is the displacement of a fluid particle in the orifice, together with Eq. (9) and assumption (7), it can be shown that Eq. (18) indeed describes the motion of some analogous mechanical system whose resistance is amplitude-dependent.

There are several interesting aspects of the resonator problem that became clearer through the use of the fluid-mechanical approach. It has been shown that energy losses in the resonator are due to viscous dissipation at the walls as well as due to the "conversion" of the momentum of the jets into vortex rings that dissipate into turbulence. The frictional wall losses are constant while the jet losses are amplitude-dependent and they dominate the constant resistance term during high-amplitude oscillations. The theoretical predictions of the resistance as function of the amplitude are presented in Fig. 2, where they are compared with available experimental data⁽⁷⁾ as well as other theoretical predictions⁽¹¹⁾. Figure 2 shows that the theoretical predictions are in good agreement, particularly at high amplitudes, with the experimental data.

The disagreement between the theoretical predictions and the experimental findings at intermediate values of velocity amplitude must be caused

* It should be noted that ξ is a Lagrangian quantity while u is an Eulerian quantity. The difference between these quantities is of second-order and will produce particular solutions that are proportional to higher harmonics. The latter are not considered in this analysis.

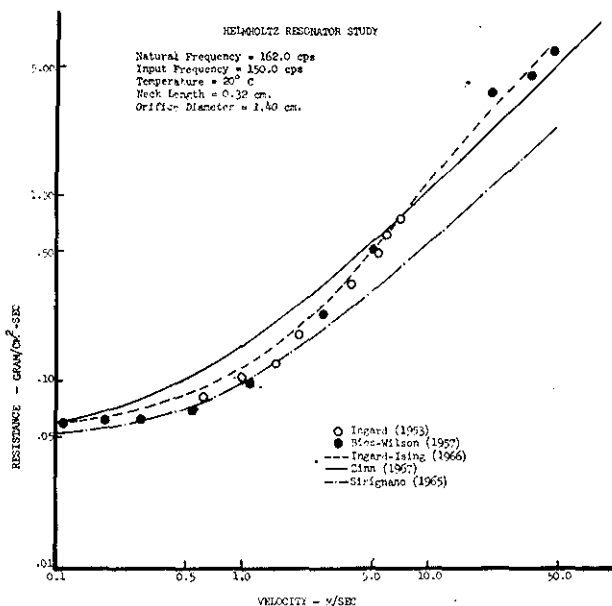


FIGURE 2. RESONATOR RESISTANCE

by the change in character of the resonator flow in this region of operation; a change that is not accounted for in this study. The constant resistance observed at low amplitudes is primarily due to viscous forces and its value is determined by the expression given in Eq. (16). It should be emphasized that this empirical expression was determined from tests conducted at low amplitudes and in reality it may change as the amplitude of the oscillation increases. Due to the lack of a more accurate expression and for the sake of preserving continuity between the results predicted by linear and nonlinear theories it was decided to use this expression in the present study

In addition to predicting the resistance this analysis also suggests that the effective value of the product of the orifice's length and the cavity's volume be used instead of the effective length used to date. This effective product could be determined experimentally in the same manner as the effective length has been determined to date.

Using the expressions derived in previous sections it can be shown that the resonator's admittance (say Y) can be expressed as

$$Y = \frac{\tilde{u}(2)}{p_1'} = \frac{\alpha(2)}{\epsilon} e^{i(\phi - \frac{\pi}{2})}$$

Plots of $\text{Re}\{Y\}$ and ϕ are given in Figs. 3 and 4. Figure 3 shows that at low amplitudes the resonator absorbs energy efficiently over a very narrow frequency range. As the amplitude increases the resonator's damping "ability" decreases while the frequency range over which it is effective increases. Figure 4 shows the dependence of the phase angle ϕ upon the velocity amplitude and the frequency. The trends shown in this figure are in qualitative agreement with available experimental observations (14).

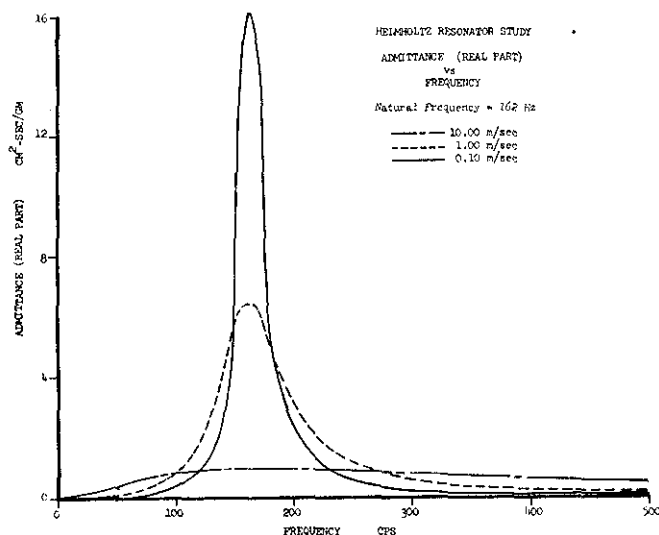


FIGURE 3. RESONATOR RESPONSE

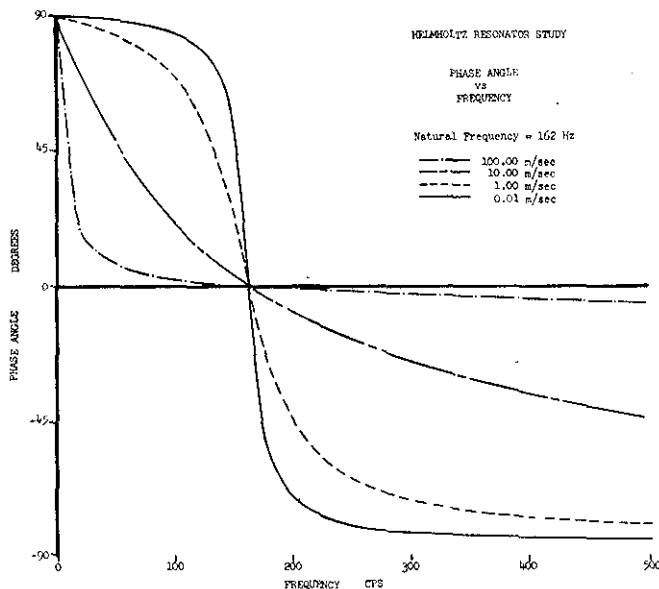


FIGURE 4. RESONATOR PHASE

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